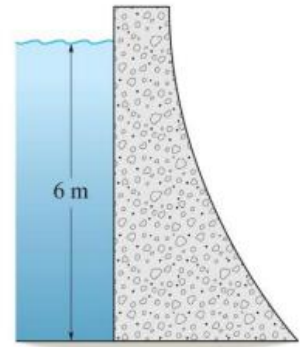


Practice Exam 4 from Dr. Koen

Practice Exam 4 – CE 2200, Dr. Koen

Section 1 – Fluid Pressure

1.1 Determine the **magnitude** of the resultant hydrostatic force acting on the dam. Width of the dam is 8 m. Density of water is 1.0 Mg/m^3 .



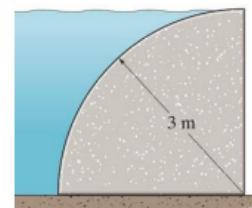
1.2 Check all the **TRUE** statements about fluid pressure problems.

- A. The center of pressure is always at the centroid of the submerged plate.
- B. The resultant force passes thru the centroid of the loading diagram.
- C. The center of pressure is the location where the resultant force strikes the plate.
- D. The center of pressure is never at the centroid of the submerged plate.

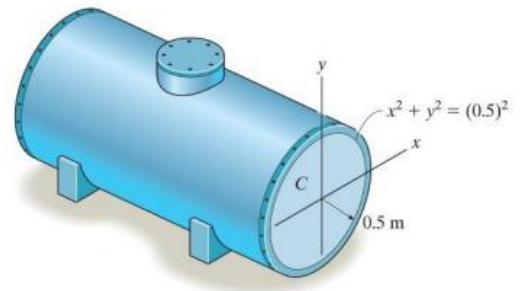
1.3 What single shape would be used to determine the **Horizontal** component of force on the side of the curved constant width plate submerged in fluid.



1.4 Determine the **Vertical** force component that the water exerts on the side of the quarter circular shaped dam **per meter of length**. Density of water is 1.0 Mg/m^3 .

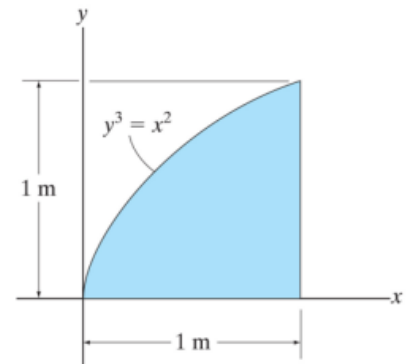


1.5 Determine the **magnitude** of the resultant hydrostatic force acting on the **flat end plate "C"** of the tank. Density of water is 1.0 Mg/m^3 .



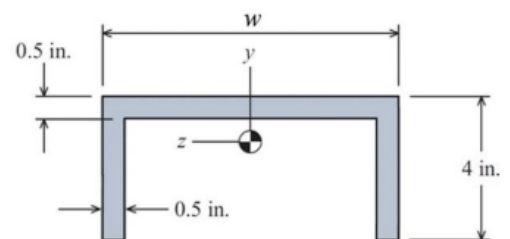
Section 2 – Moment of Inertia

2.1 Determine the moment of inertia for the shaded about the x axis.



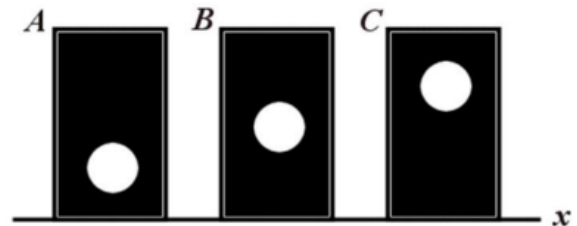
2.2 TRUE FALSE For a general area, the parallel axis theorem would be used when calculating I_y and using a horizontal differential element.

2.3 If $w = 8.50 \text{ in.}$, find the moment of inertia about the centroidal z-axis. The centroid of the shape is located 2.8468 inches above the bottom of the shape.



2.4 A general shape with an area of 20 in^2 has a moment of inertia of 106.7 in^4 with respect to the x axis. The centroid is located 2 inches above the x axis. What is the moment of inertia with respect to a centroidal x' axis?

2.5 The figure shows three areas. Each area is a rectangle with a missing circle. The three rectangles have the same dimensions, and the three missing circles have the same dimensions. Which area has the largest moment of inertia about the x axis?

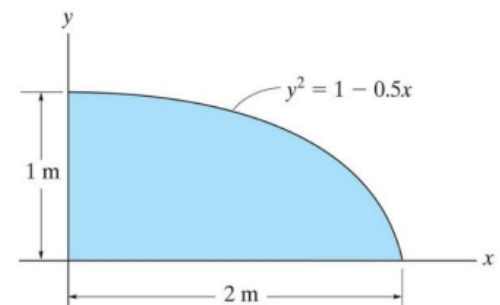


Section 3 – Product of Inertia

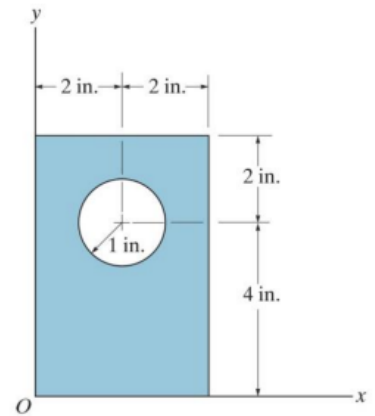
3.1 When using the parallel axis theorem to determine the product of inertia for an area about a set of non-centroidal x - y axes, $I_{xy} = \bar{I}_{x'y'} + Ad_xd_y$, the second term on the right side of the equation is:

- A. Area times the differential widths dx and dy
- B. Area times the distances between the x - y origin and the x' - y' origin.
- C. Always positive
- D. Always negative
- E. Always zero

3.2 Determine the product of inertia of the shaded area with respect to the x and y axes.

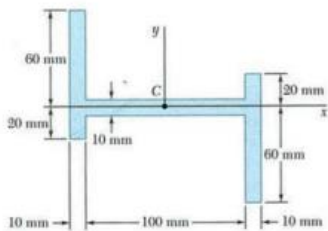


3.3 Determine the product of inertia for the shaded area with respect to the x and y axes.

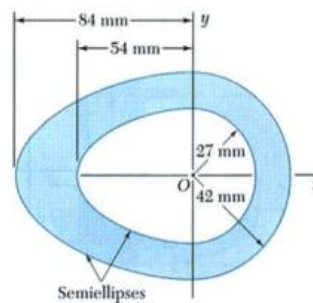


By inspection, determine the sign of I_{xy} for figures based on the coordinate systems given. Choose from the following choices: positive negative zero

3.4



3.5



Section 4 – Inclined Axis and Principle Moments of Inertia

4.1 The orientation (*relative to the xy axes shown*) of the **principal axes** with origin at the centroid for the cross sectional area would be _____ degrees.

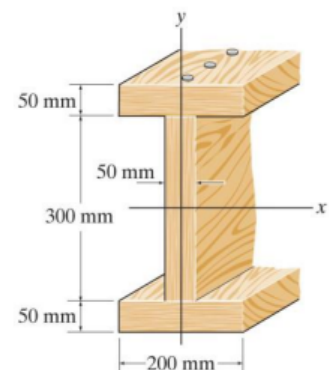


Figure: 10_FP007

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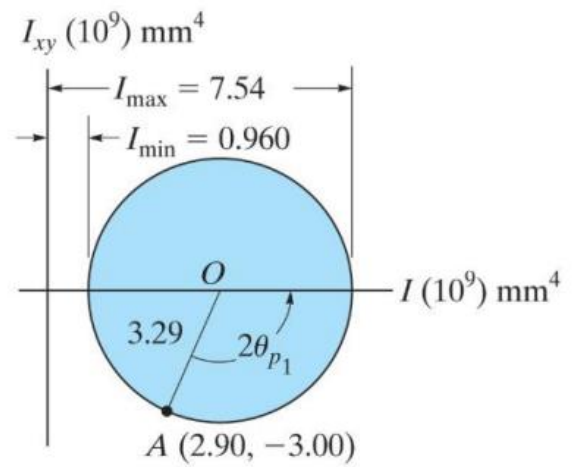
A **Mohr's circle** has the reference point **A** (I_x , I_{xy}).

The **center** of the circle is at **4.25** and the **radius** is **3.29**

4.2 $I_y = \underline{\hspace{2cm}} 10^9 \text{ mm}^4$

4.3 $I_{\min} = \underline{\hspace{2cm}} 10^9 \text{ mm}^4$

4.4 $2\theta = \underline{\hspace{2cm}}^\circ$



4.5 On the **actual cross section**, the **major** principal axis would be shown rotated degrees from the positive x axis.

Answers

- 1.1 1413 kN
- 1.2 B and C are true
- 1.3 Trapezoid
- 1.4 18.9 kN
- 1.5 3.85 kN

-
- 2.1 0.111 m^4
 - 2.2 TRUE
 - 2.3 11.34 in^4
 - 2.4 26.7 in^4
 - 2.5 A

3.1 B

3.2 0.333 m^4

3.3 119 in^4

3.4 negative

3.5 zero

4.1 0 degrees

4.2 5.6

4.3 .960

4.4 114.2 degrees

4.5 57.1 degrees CCW